# Remote Controller with Fail-Safe Update

Software Classification

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|---|---|
| Reference | TBD **[TBD-1]** |
| Issue / revision | Issue 1, Revision 0 — DRAFT |
| Date | 10 October 2026 |
| Author | Andrew Toma |
| Open items | 2 TBD (App. A) |

**Document change record**

| Issue | Rev. | Date | Change |
|---|---|---|---|
| 1 | 0 | 2026-10-10 | First draft |

## 1.0 Purpose and Scope

This document establishes the software classification of each software item, and so the rigour each item requires. It is the formal record of the software classification determinations [1, Sec. 3.5.2, p. 25].

The evaluated software items encompass the System software and its enabling systems, as defined in the Concept of Operations [4, Sec. 1.2.2]. The System models an uncrewed spacecraft whose primary mission objectives directly depend on this software; a failure would cost those objectives.

## 2.0 Documents

[1] NASA, *NASA Software Engineering Requirements*, NPR 7150.2D.
[2] NASA, *Software Assurance and Software Safety Standard*, NASA-STD-8739.8B, 2022.
[3] NASA, *NASA Software Engineering Handbook*, NASA-HDBK-2203, Version D, page "SWE-020 – Software Classification".
[4] *Remote Controller with Fail-Safe Update — Concept of Operations*, Issue 1, Revision 0, Draft.

## 3.0 Classification

Each item takes the highest class whose definition applies [1, Sec. 3.5.1, p. 24].

| Software item | Role | Class |
|---|---|---|
| Controller firmware | FLIGHT | B |
| Onboard computer software | FLIGHT | B |
| Ground software in the mission operations center | GROUND | B |
| Bench test tools | TEST | C |
| Firmware build tools | BUILD | D |

**Legend: role**

| | |
|---|---|
| **FLIGHT** | Operates the spacecraft from orbit. |
| **GROUND** | Operates the spacecraft from the ground. |
| **TEST** | Tests the System; never operates the spacecraft. |
| **BUILD** | Builds the System; never operates the spacecraft. |
| **Class** | As defined in [1, App. D, pp. 79–86]. |

### 3.1 Rationale

**Controller firmware: Class B.** Class B covers flight software "required to operate the vehicle or space asset ... such as commanding of the vehicle or asset" [1, App. D, p. 81]; its examples include "fault protection" [1, App. D, p. 81]. The controller runs commands and returns to working firmware on its own when new firmware fails.

**Onboard computer software: Class B.** The same definition applies. It passes every command and firmware image to the controller, watches it, and restores it when its firmware cannot run.

**Ground software: Class B.** Class B examples include "command and control ground systems" [1, App. D, p. 81]. The ground software prepares and checks every command and firmware image sent to the spacecraft.

**Bench test tools: Class C.** Class C covers "software used for the testing of space assets" [1, App. D, p. 82], and Class B excludes systems "used in testing Class B systems" in a development environment [1, App. D, p. 81].

**Firmware build tools: Class D.** Class D covers "ground software tools that support engineering development" [1, App. D, p. 83].

## 4.0 Safety-Critical Designation

Some software is safety-critical: the parts that can cause a hazard, or that exist to prevent, detect or handle one [2, Sec. 4.2, p. 20]. Safety-critical software must meet extra requirements on top of its class, for example test coverage of every decision (MC/DC) [1, Sec. 3.7.4, pp. 26–27].

Which items are safety-critical can only be decided once the hazards are known, in the hazard analysis [**TBD-2**].

## 5.0 Tailoring and Review

- **Joint agreement and single-developer tailoring:** Software engineering and software assurance are expected to agree on each classification [1, Sec. 3.5.1 Note, p. 25]. One engineer develops the System, with no independent software assurance team (ConOps constraint C-3); the classification therefore rests with the author alone.
- **Life-cycle re-evaluation:** The classification is revisited at each later phase [3, Sec. 3.5.1]. If an item's class changes, the plans are updated to match [1, Sec. 2.2.8, p. 19].

## A Open Items

The values below are not yet known.

**TBD-1** Project name and document reference.

**TBD-2** Safety-critical designation of each software item, from the hazard analysis.

# B Requirements Mapping Matrix

Every requirement of NPR 7150.2D marked for an item's class is complied with [1, App. C, pp. 56–78], except those tailored out below [1, Sec. 3.1.12, p. 22]. The tailoring was decided by the author on 10 October 2026 and is revisited at each later phase.

| SWE | Requirement | Reason |
|---|---|---|
| 141 | Software IV&V for NASA project categories | NOT NASA |
| 143 | Architecture review for NASA project categories | NOT NASA |
| 174 | Planning data to the Center measurement repository | NOT NASA |
| 148 | Reuse candidates to NASA's catalogue | NOT NASA |
| 094 | Measurement data to the sponsoring organisation | NOT NASA |
| 039 | Status reports from the software developer | NO SUPPLIER |
| 040 | Products and records from the developer to NASA | NO SUPPLIER |
| 042 | NASA access to the developer's source code | NO SUPPLIER |
| 046 | Software schedule from the developer | NO SUPPLIER |
| 206 | NASA access to the developer's models and simulations | NO SUPPLIER |
| 045 | Joint NASA/developer audits | NO SUPPLIER |
| 131 | IV&V plan | NO IV&V |
| 178 | IV&V access to development artifacts | NO IV&V |
| 179 | Responses to IV&V issues and risks | NO IV&V |
| 032 | CMMI-DEV rated organisation | NO RATING |
| 022 | Software assurance and safety plan per NASA-STD-8739.8 | NO ASSURANCE |
| 205 | Safety-critical determination with the SMA organisation | NO ASSURANCE |
| 015 | Two cost estimate models | NO BUDGET |
| 151 | Conditions on the cost estimate | NO BUDGET |
| 070 | Accredited models and simulations for flight qualification | BENCH ONLY |

**Legend: reason**

**NOT NASA** The System is not a NASA program or project; there is no NASA Center, catalogue or sponsoring organisation.

**NO SUPPLIER** The engineer develops all software; there is no separate developer or supplier.

**NO IV&V** Independent verification and validation is not performed.

**NO RATING** There is no organisation to hold a CMMI rating.

**NO ASSURANCE** There is no independent software assurance or SMA organisation (ConOps constraint C-3); the author decides alone.

**NO BUDGET** The project has no development budget to estimate.

**BENCH ONLY** The System is demonstrated on a bench, not qualified for flight (ConOps constraint C-4).